

Problem

We can modify binary search algorithm to perform division of two numbers, by defining range $[0, \text{infinity}]$ which serves as initial low and high for the binary search algorithm. Now we need to find a mid that satisfies $x/y = \text{mid}$ or $x * \text{mid}$ for given two numbers x and y . Based on the comparison result based on x and $y * \text{mid}$, we either update low, update high or return mid. Division with Binary Search

Submissions

1.If $y * \text{mid}$ almost equal to x , we return mid. 2. If $y * \text{mid}$ is less than x , we update low to mid 3. If $y * \text{mid}$ is more than x , we update high to mid We need to care about division by zero and sign of the result etc. Input: one line of input should contain two numbers separated by space. Output: should print division of the numbers as a result.

Leaderboard

Input Format

one line of input should contain two numbers separated by space. Input1: 22 7

Constraints

$x, y < \text{infinity}$

Output Format

should print division of the numbers as a result

Discussions

Sample Input 0

22 2

Sample Output 0

11

Sample Input 1

12 6

Sample Output 1

2

[Change Theme](#)

Language Python 3



```
1 # Enter your code here. Read input from STDIN.
2 def divide(x, y):
3     if y == 0:
4         return "Division by zero is not allowed"
5
6     # Handle sign
7     negative = (x < 0) ^ (y < 0)
8
9     dividend = abs(x)
10    divisor = abs(y)
11
12    low = 0
13    high = dividend
14    ans = 0
15
16    while low <= high:
17        mid = (low + high) // 2
18
19        if mid * divisor == dividend:
20            ans = mid
21            break
22        elif mid * divisor < dividend:
23            ans = mid
24            low = mid + 1
25        else:
```

Line: 36 Col: 1



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0



Test case 1

Compiler Message

Success

